PARISHRAM 2.0

CLASS 12TH 2026

Botany-Sexual Reproduction in Flowering Plants

BIOLOGY

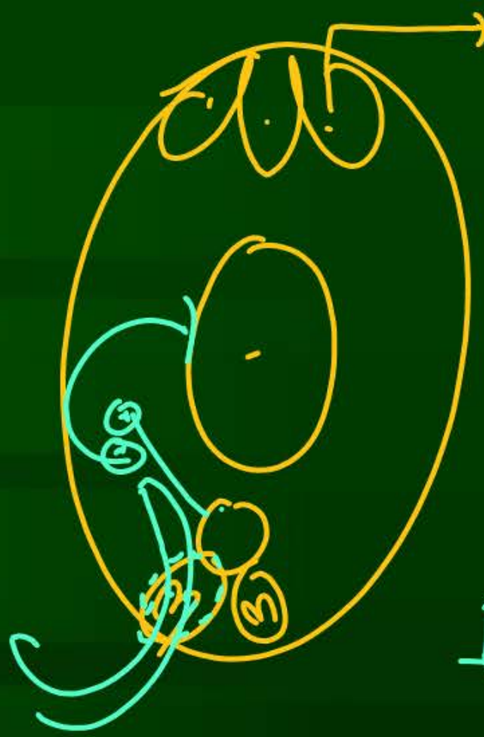
LECTURE-8

BY-AYUSHI AGARWAL MA'AM



TOPICS TO BE COVERED

1. Post fertilization changes
2.
3.
4. Type option here



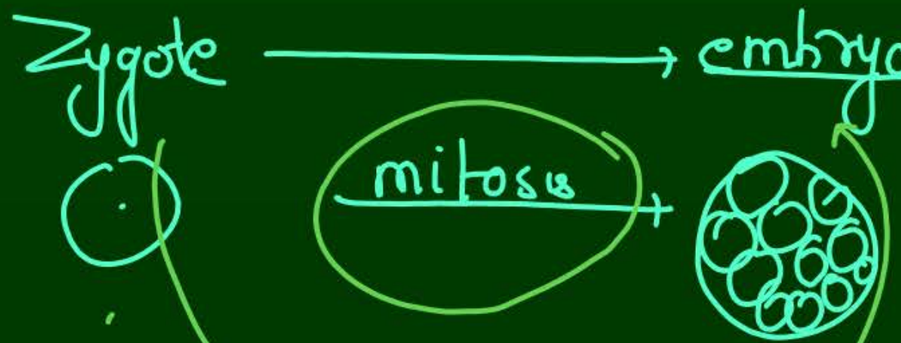
Syngamy → Zygote

Triple fusion → PEC

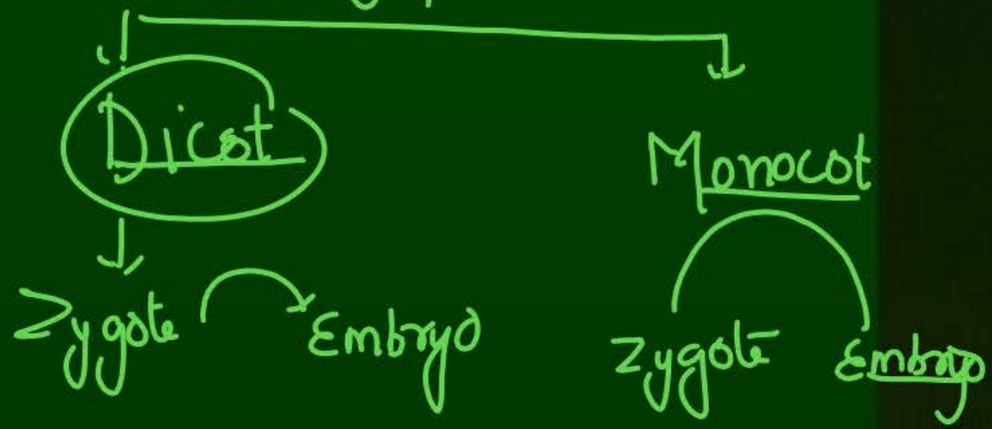
Double fertilization.

Embryogenesis

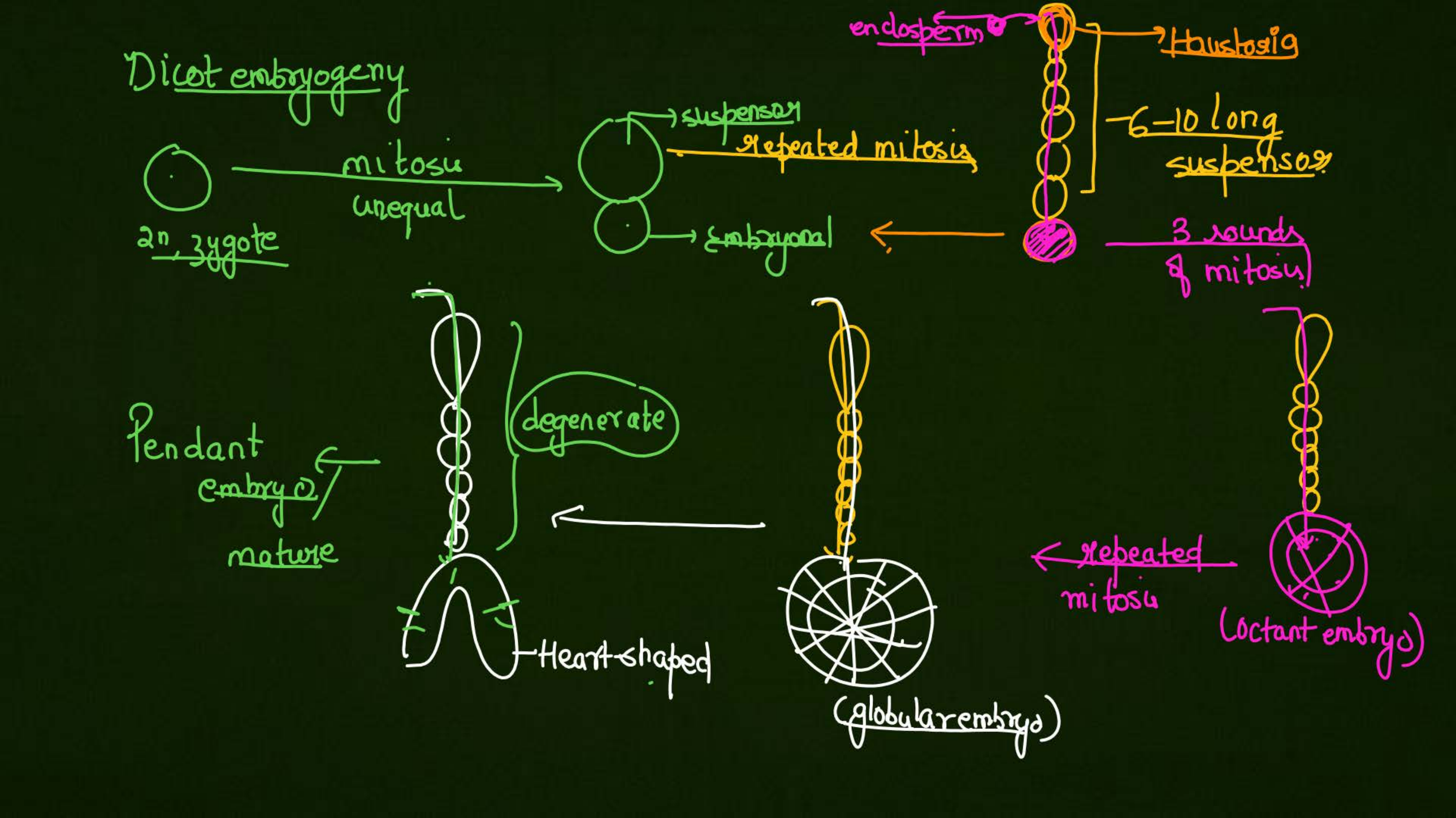
Endosperm → food rich mass of tissue.

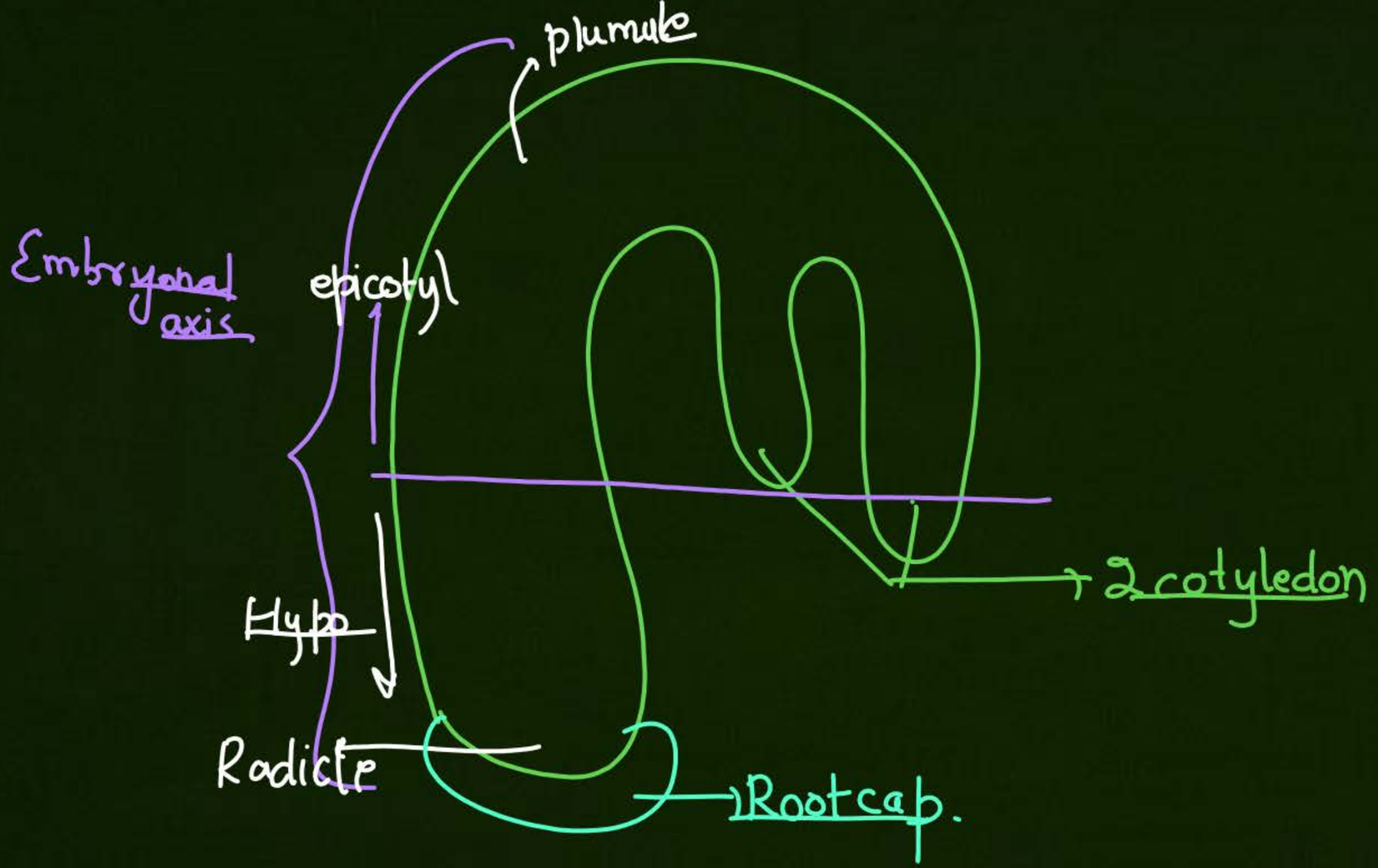


Angiosperm



→ formation of embryo from zygote after mitosis


Dicot embryogeny
2n zygote
mitosis
unequal
suspensor
repeated mitosis
embryonal
endosperm
Haustoria
6–10 long suspensor
3 rounds of mitosis
(globular embryo)
repeated mitosis
(octant embryo)
degenerate
Heart-shaped
Pendant embryo mature



Embryo

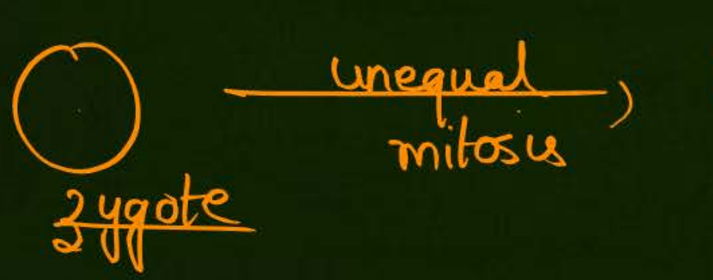
- Cotyledon (2/1)
- Embryonal axis
  - epicotyl → Plumule → Shoot
  - Hypo → Radicle → Root.

Monocot embryogeny

1. Initial stage of dev. is same in both monocot & dicot.

2. Zygote —mitosis→ Embryo

(source of nutrition is not endosperm)

3. 

, X, because there is no need to extract nutrients from endosperm

∴ monocot seeds - endosperm is intact

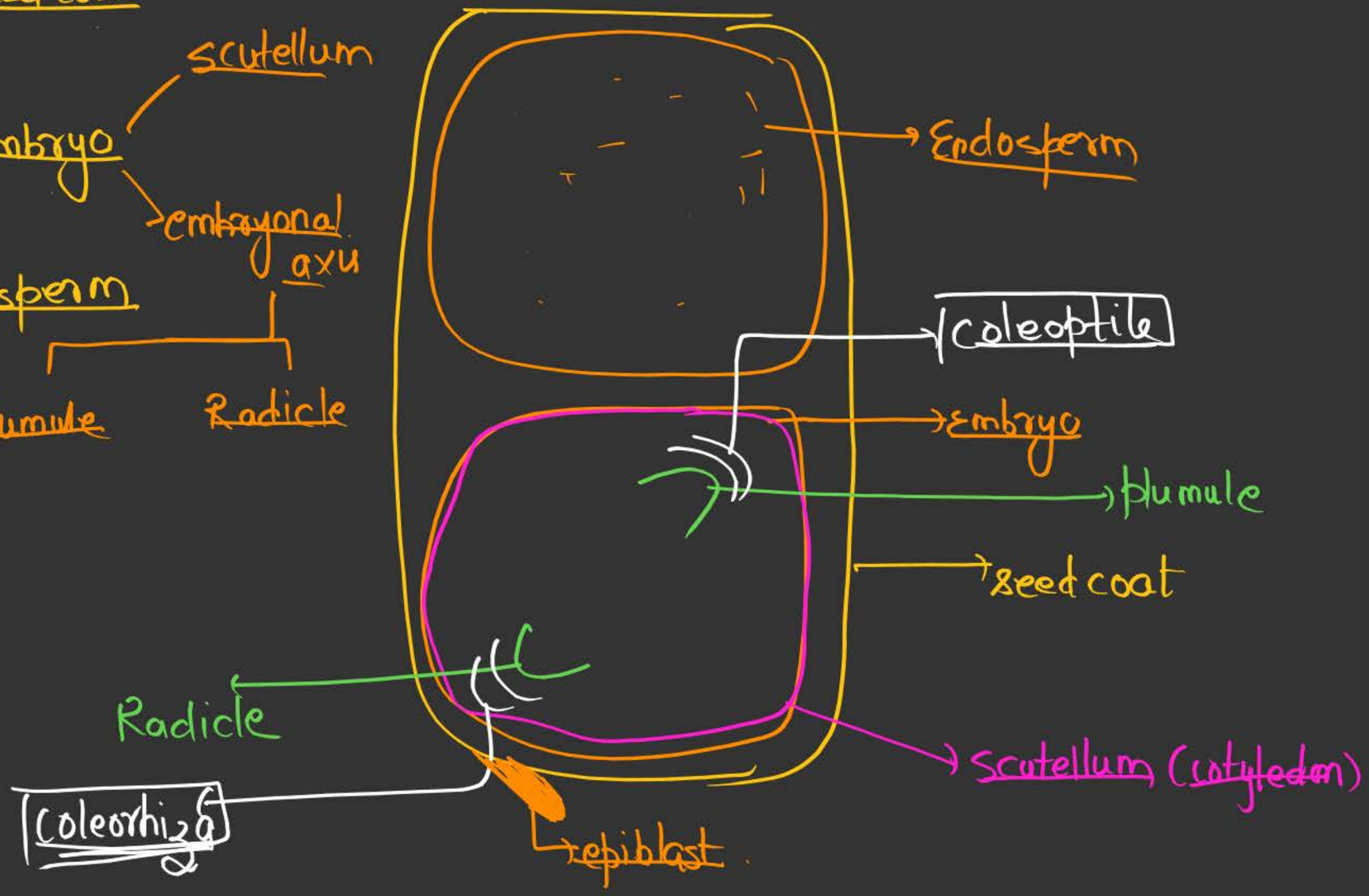
(4.) Mature monocot embryo

- Embryonal axis
- cotyledon
  - single → scutellum
  - (Note:- 2nd cotyledon germinate → epiblast)

Monocot seed

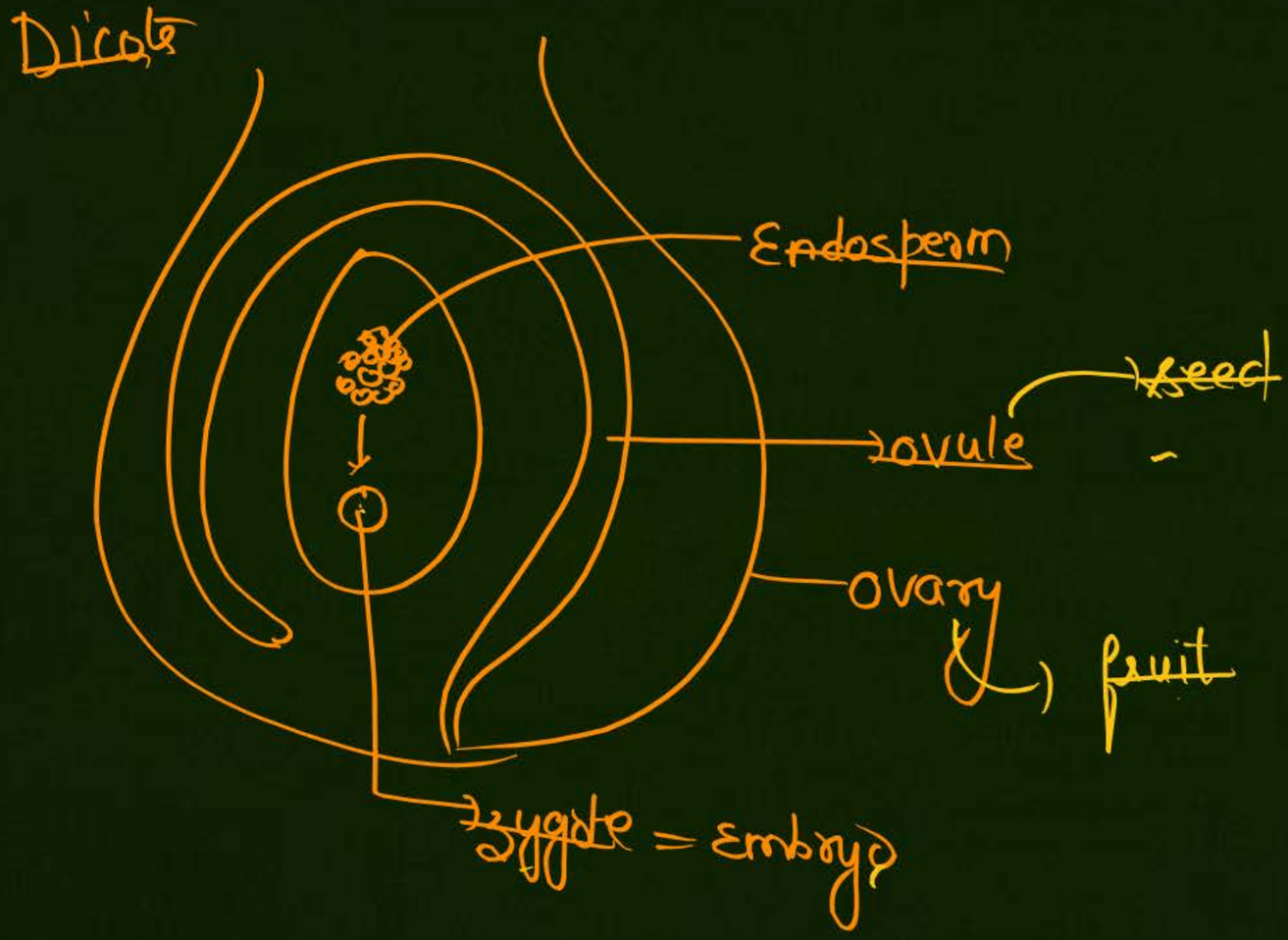
(endospermic)

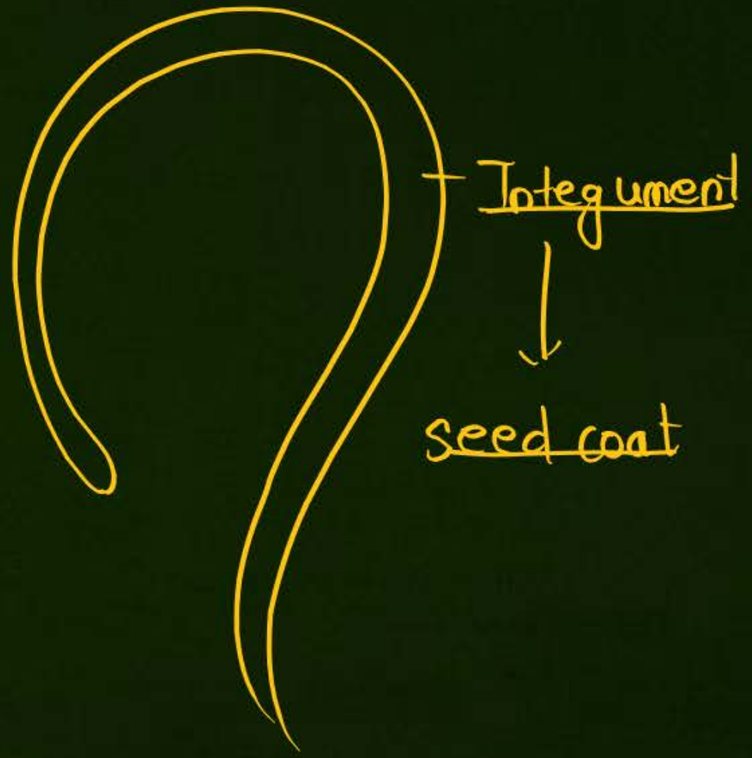
- seed coat
- embryo
  - scutellum
  - embryonal axu
    - Plumule
    - Radicle
- endosperm

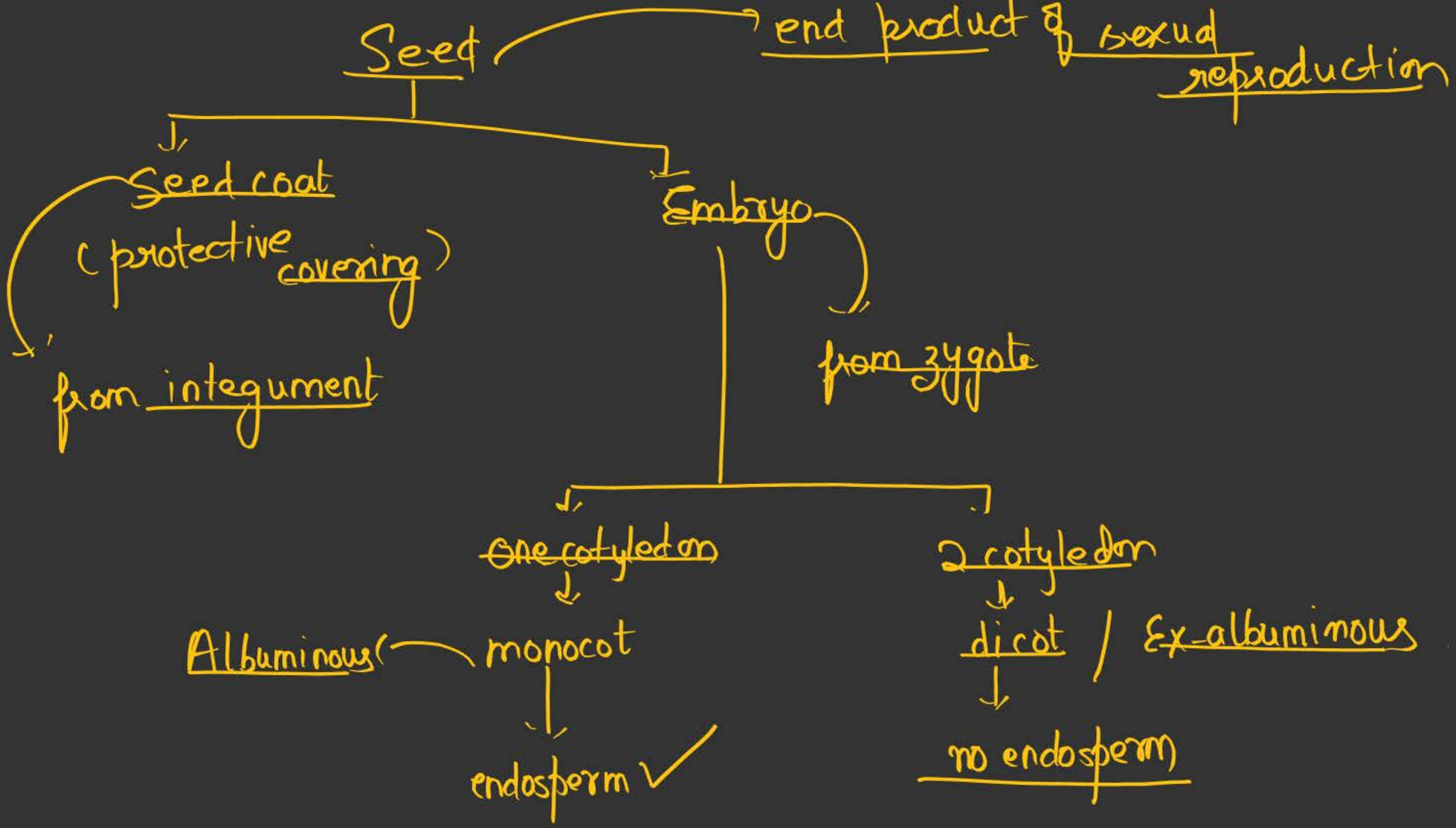


Fruit and seed

Seed → fertilized ovule






Seed
end product of sexual reproduction
Seed coat
(protective covering)
from integument
Embryo
from zygote
one cotyledon
2 cotyledon
Albuminous
monocot
endosperm
dicot / Ex-albuminous
no endosperm

Seed is enclosed within fruit ⟶ fertilized ovary

True
I

only from ovary

ovary

false - eg apple, pear, peach.

ovary + Thalamus — fruit

Parthenocarpic — without fertilization from ovary

Banana

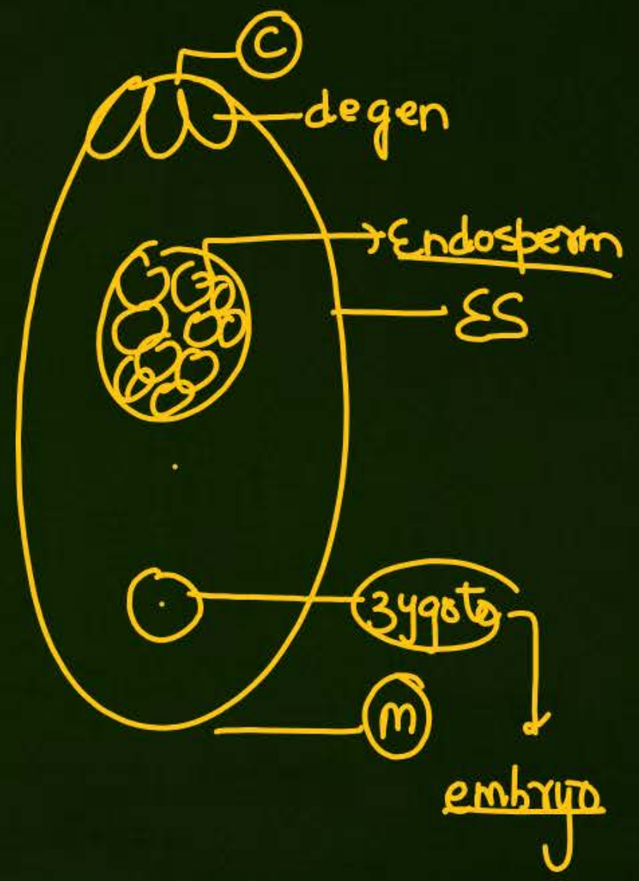
Grapes, ~~watermelon~~) seedless

1.4.2 Embryo

Embryo develops at the micropylar end of the embryo sac where the zygote is situated. Most zygotes divide only after certain amount of endosperm is formed. This is an adaptation to provide assured nutrition to the developing embryo. Though the seeds differ greatly, the early stages of embryo development **(embryogeny)** are similar in both monocotyledons and dicotyledons. Figure 1.13 depicts the stages of embryogeny in a dicotyledonous embryo. The zygote gives rise to the **proembryo** and subsequently to the **globular, heart-shaped** and **mature embryo**.

A typical dicotyledonous embryo (Figure 1.14a), consists of an **embryonal axis** and two **cotyledons.** The portion of embryonal axis above the level of cotyledons is the **epicotyl**, which terminates with the **plumule** or stem tip. The cylindrical portion below the level of cotyledons is **hypocotyl** that terminates at its lower end in the **radicle** or **root tip**. The root tip is covered with a **root cap.**

Embryos of monocotyledons (Figure 1.14 b) possess only one cotyledon. In the grass family the cotyledon is called **scutellum** that is situated towards one side (lateral) of the embryonal axis. At its lower end, the embryonal axis has the



radical and root cap enclosed in an undifferentiated sheath called **coleorrhiza.** The portion of the embryonal axis above the level of attachment of scutellum is the epicotyl. Epicotyl has a shoot apex and a few leaf primordia enclosed in a hollow foliar structure, the **coleoptile**.

Soak a few seeds in water (say of wheat, maize, peas, chickpeas, ground nut) overnight. Then split the seeds and observe the various parts of the embryo and the seed.

1.4.3 Seed

In angiosperms, the seed is the final product of sexual reproduction. It is often described as a fertilised ovule. Seeds are formed inside fruits. A seed typically consists of seed coat(s), cotyledon(s) and an embryo axis. The cotyledons (Figure 1.15a) of the embryo are simple structures, generally thick and swollen due to storage of food reserves (as in legumes). Mature seeds may be **non-albuminous** or **ex-albuminous**. Non-albuminous seeds have no residual endosperm as it is completely consumed during embryo development (e.g., pea, groundnut). Albuminous seeds retain a part of endosperm as it is not completely used up during embryo development (e.g., wheat, maize, barley, castor). Occasionally, in some seeds such as black pepper and beet, remnants of nucellus are also persistent. This residual, persistent nucellus is the **perisperm**.

Perispermic seed

Nucellus → degenerate

except → Black pepper
beet

↓

seed (nucellus)

Integuments of ovules harden as tough protective seed coats (Figure 1.15a). The micropyle remains as a small pore in the seed coat. This facilitates entry of oxygen and water into the seed during germination. As the seed matures, its water content is reduced and seeds become relatively dry (10-15 per cent moisture by mass). The general metabolic activity of the embryo slows down. The embryo may enter a state of inactivity called **dormancy,** or if favourable conditions are available (adequate moisture, oxygen and suitable temperature), they germinate.

As ovules mature into seeds, the ovary develops into a fruit, i.e., the transformation of ovules into seeds and ovary into fruit proceeds simultaneously. The wall of the ovary develops into the wall of fruit called **pericarp**. The fruits may be fleshy as in guava, orange, mango, etc., or may be dry, as in groundnut, and mustard, etc. Many fruits have evolved mechanisms for dispersal of seeds. Recall the classification of fruits and their dispersal mechanisms that you have studied in an earlier class. *Is there any relationship between number of ovules in an ovary and the number of seeds present in a fruit?*

In most plants, by the time the fruit develops from the ovary, other floral parts degenerate and fall off. However, in a few species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called **false fruits** (Figure 1.15b). Most fruits however develop only from the ovary and are called **true fruits**. Although in most

1.5 Apomixis and Polyembryony

Although seeds, in general are the products of fertilisation, a few flowering plants such as some species of *Asteraceae* and grasses, have evolved a special mechanism, to produce seeds without fertilisation, called **apomixis.** *What is fruit production without fertilisation called?* Thus, apomixis is a form of asexual reproduction that mimics sexual reproduction. There are several ways of development of apomictic seeds. In some species, the diploid egg cell is formed without reduction division and develops into the embryo without fertilisation. More often, as in many *Citrus* and *Mango*

varieties some of the nucellar cells surrounding the embryo sac start dividing, protrude into the embryo sac and develop into the embryos. In such species each ovule contains many embryos. Occurrence of more than one embryo in a seed is referred to as **polyembryony**. *Take out*

QUESTION

How many wall layers does anther has?

1. One
2. Two
3. Three
4. Four



THANK YOU